

Parameter	Symbol	Value	Unit	Note
Radiant power	Φ	1-4	W	From OGS specs.
Diameter of limiting aperture	D_f	3.5	mm	From [REF] Table 3.
Period of exposure	t	10	s	From [REF] Table 1.
Beam divergence	ϕ	3	μrad	From OGS specs.
Beam diameter	a	70	cm	From OGS specs.

NOHD Calculations

$$\begin{aligned}
\theta &= 1.22 \cdot \frac{\lambda}{D} \\
&= 1.22 \cdot \frac{1.55 \mu\text{m}}{0.7 \text{ m}} \\
&= 2.7 \mu\text{rad}
\end{aligned}$$

$$\begin{aligned}
\text{MPE}(H) &= 1.8 \cdot t^{0.75} \frac{\text{mJ}}{\text{cm}^2} \\
&= 1.8 \cdot 10^{0.75} \frac{\text{mJ}}{\text{cm}^2} \\
&= 10.12 \frac{\text{mJ}}{\text{cm}^2}
\end{aligned}$$

$$\begin{aligned}
\text{MPE}(E) &= \frac{\text{MPE}(H)}{t} \\
&= \frac{10.12 \frac{\text{mJ}}{\text{cm}^2}}{10 \text{ s}} \\
&= 1.012 \frac{\text{mW}}{\text{cm}^2}
\end{aligned}$$

$$\begin{aligned}
D_{\text{safe}} &= \sqrt{\frac{4\Phi}{\pi \text{MPE}}} \\
&= \sqrt{\frac{4 \cdot [1; 4] \text{ W}}{\pi \cdot 1.0122 \frac{\text{mW}}{\text{cm}^2}}} \\
&= [35.46; 70.93] \text{ cm}
\end{aligned}$$

$$\begin{aligned}
\text{NOHD} &= \frac{1}{\phi} \cdot \left[\left(\frac{4\Phi}{\pi \text{MPE}} \right) - a^2 \right]^{1/2} \\
&= \frac{1}{3 \mu\text{rad}} \cdot \left[\left(\frac{4 \cdot [1; 4] \text{ W}}{\pi \cdot 1.0122 \text{ mW/cm}^2} \right) - (70 \text{ cm})^2 \right]^{1/2} \\
&= [0; 38.235] \text{ km}
\end{aligned}$$

$$\begin{aligned}
NOHD &\approx \frac{1}{\phi} \cdot \left(\frac{1.27\Phi}{\text{MPE}} \right)^{\frac{1}{2}} \\
&\approx \frac{1}{1 \text{ mrad}} \cdot \left(\frac{1.27 \cdot [1; 4] \text{ W}}{1.0122 \frac{\text{mW}}{\text{cm}^2}} \right) \\
&\approx [354; 708] \text{ m}
\end{aligned}$$

$$NOHD_{\text{SMF}} = \frac{\omega_0}{\lambda} \left(\frac{\pi\Phi}{2 \cdot \text{MPE}} \right)^{1/2}$$

$$\begin{aligned}
E &= \frac{\Phi}{\pi(D_f/2)^2} \\
&= \frac{[1; 4] \text{ W}}{\pi(0.35 \text{ cm}/2)^2} \\
&= [10.39; 41.57] \frac{\text{W}}{\text{cm}^2}
\end{aligned}$$

$$\begin{aligned}
\text{OD}_\lambda &= \log \left(\frac{E}{\text{MPE}} \right) \\
&= \log \left(\frac{[10.39; 41.57] \text{ W/cm}^2}{1.0122 \text{ mW/cm}^2} \right) \\
&= [4.01; 4.6]
\end{aligned}$$

1 References

?? https://d3qi0qp55mx5f5.cloudfront.net/researchsafety/docs/Laser_Safety_Calculations.pdf?mtime=1610127144